DECCAN EDUCATION SOCIETY - Since 1884

DES PUNE UNIVERSITY

School of Engineering and Technology

Presents

Nav-उन्मेष 2026

Circuit Designing Competition - Problem Statements

NOTE: Microcontrollers are strictly prohibited for implementing any of the below statements.

1. **Automatic Battery Charging Cut-Off Circuit**

Design a circuit that monitors battery voltage and disconnects the charger at a preset level using a voltage divider and comparator (LM358/LM393). The output should drive a relay/MOSFET, with potentiometer-adjustable cutoff and LEDs for charging/full indication.

2. **Programmable Frequency Divider**

Design a circuit that divides a master clock using a binary ripple counter (74HC93/4040) to generate ÷2, ÷4, ÷8, and ÷16 outputs. Division selection must be via switches, while all outputs remain stable and available simultaneously.

3. **Sequential Code-Based Digital Lock System**

Design a 4-switch sequence digital lock that opens only when the correct order is entered. Use shift registers/latch chains and logic gates for sequence validation, with manual reset after wrong attempts.

4. **Window Comparator Safe Range Indicator**

Design a circuit that indicates under-voltage, normal, and over-voltage using two comparators (LM358/LM393). Upper and lower thresholds must be independently adjustable using resistor networks/potentiometers with LED indicators.

5. **Runway Light Sequential Controller**

Design a runway LED sequencing system using a 555 Timer (clock) and 4017 Decade Counter. LED sequence speed must be adjustable via potentiometer and operate smoothly in order.

6. **Advanced Digital Lock with Security Penalty Logic**

Design a sequence-based digital lock that triggers a 10-second lockout after a wrong attempt and activates a permanent alarm after three failures. The system must require a manual reset to restore operation.

7. **Electronic Quiz Buzzer Lockout System**

Design a quiz buzzer circuit that detects the first button press and locks out others using SR/D flip-flops. The winning participant should be indicated with an LED, with a manual reset for the next round.

8. **Automatic Doorbell Pulse Controller**

Design a doorbell circuit using a 555 Timer in monostable mode to produce a fixed-duration ring per button press. The system must ignore long presses and automatically reset for the next trigger.

9. **Emergency Vehicle Traffic Signal Override System**

Design a traffic signal controller using a 555 Timer and 4017 counter for normal sequencing. A manual override should instantly give GREEN to the selected lane, then resume normal operation without conflicts.